ASHOK GURUNG, Ph.D.

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Summary

Computational physicist with 6+ years of experience in multiscale and multiphysics modeling of electro-chemo-mechanical phenomena in electroactive materials. Expertise spans phase-field modeling, finite element methods (MOOSE and COMSOL), molecular dynamics, and density functional theory (VASP) for electroactive and functional ceramics. Experienced in integrating simulations with experiments, leveraging Python-based automation, high-performance computing (HPC), and data-driven/AI-assisted approaches for predictive materials design.

Education

University of Connecticut (UConn), CT, US **Ph.D. in Physics, Dec 2024**
Dissertation: *Predictive Multiscale Modeling of Dielectric and Electromechanical Properties in Electroactive Materials*
Advisor: Prof. Serge M. Nakhmanson

Wright State University (WSU), OH, US **M.S. in Physics, July 2018**
Thesis: *Molecular Dynamics Study of Hydrogen Trapping and Helium Clustering in Tungsten*
Advisor: Prof. Amit Sharma

Professional Experience

Postdoctoral Research Associate **Jun. 2025 – Present**
Washington State University *Pullman, WA, USA*
Advisors: Prof. Sinisa Mesarovic and Prof. Scott Beckman

Postdoctoral Research Associate **Sept. 2024 – Jun. 2025**
Rochester Institute of Technology *Rochester, NY, USA*
Advisor: Prof. Howard Tu

Research Assistant **Jan 2021 – Aug 2024**
University of Connecticut *Storrs, CT, USA*
Advisor: Prof. Serge Nakhmanson

Teaching Assistant **Aug 2018 – Dec 2021**
University of Connecticut *Storrs, CT, USA*
Advisor: Prof. Serge Nakhmanson

Teaching Assistant **Aug 2016 – Jun 2018**
Wright State University *Dayton, OH, USA*
Advisor: Prof. Amit Sharma

Technical Skills

Multiphysics and Continuum-Scale Modeling: Phase-field modeling coupled with diffusion, electrostatics, and mechanics; finite element modeling (FEM) using MOOSE and COMSOL

Atomistic Modeling: Classical molecular dynamics and density functional theory (DFT) using VASP

Data and Machine Learning: Data mining, Python- and Shell-based automation, data-driven materials modeling, machine learning

Programming and Software: Python, C/C++, MATLAB, shell scripting, LaTeX, and GitHub

Operating Systems and HPC: Ubuntu/Linux, high-performance computing clusters, and Windows

Publications

- **A. Gurung**, M. Fatin Ishtiyaq, T. Sahoo, S. M. Nakhmanson, S. K. Nayak, S. P. Alpay, and P. E. Janolin, *"Electrostrictive Biomaterials"* (in preparation 2026).
- **A. Gurung**, S. Nayak, R. Esteban Goetz, *"Quantum Matters: From Field to Life"*, Book, Taylor and Francis, Under Publication, 2026.
- A. N. Bage, J. M. Vazquez Mercado, F. D. Cúñez, **A. Gurung**, S. Li and H. Tu, *"Dendrite Suppression by Detouring Li Transport within a Mechanically Anisotropic Solid Electrolyte"*, Nano Letters (2026) .
- R. Elawadly, J. M. Vazquez Mercado, F. D. Cúñez, **A. Gurung**, I. A. Shozib, Q. H. Tu, *Tuning Effective Mechanical Properties of Catholyte to Enhance Cathode Performance in Solid-state Batteries*,2025, http://dx.doi.org/10.2139/ssrn.5534041
- **A. Gurung**, J. M. Mangeri, M. Fatin Ishtiyaq, S. P. Alpay, and S. Nakhmanson, *"Extrinsic Dielectric Response due to Domain Wall Motion in Ferroelectric $BaTiO_3$"*, Computational Materials Today. 5, 100016 (2025).
- **A. Gurung**, J. M. Mangeri, A. M. Hagerstrom, N. D. Orloff, S. P. Alpay, and S. Nakhmanson, *"Modeling Structure-Properties Relations in Compositionally Disordered Relaxor Dielectrics at the Nanoscale"*, J. Appl. Phys. 134, 104102 (2023).
- **A. Gurung***, D. Trujillo* (equal contribution), J. Yu, S. K. Nayak, S. P. Alpay, and P-E. Janolin, *"Data-driven Methods for Discovery of Next-generation Electrostrictive Materials"*, npj Comp. Mater. 8, 251 (2022).
- **A. Gurung**, *"Molecular Dynamics Study of Hydrogen Trapping and Helium Clustering in Tungsten"*, master's thesis, Wright State University, 2018.

Professional Experience

Phase field modeling of sintering of ceramic materials **Jun 2025 – Present**
Postdoctoral Research Associate, School of Mechanical and Materials Engineering, WSU *Pullman, WA*

- Designed and implemented a coupled phase-field and concentration-field formulation to model sintering mechanisms, including surface, bulk, and interface diffusion, enabling quantitative analysis of microstructural evolution.
- Tested and integrated mechanical modulus effects into the sintering model to evaluate behavior under various experimental conditions, providing insights that guided experimental parameter selection.
- Used Thermo-Calc and pycalphad to compute temperature-dependent free energies and integrate thermodynamic inputs into a robust numerical framework for modeling high-temperature joining of dissimilar materials.

NSF-Engine Workforce Development Summer Camp **Apr 2025 – May 2025**
Postdoctoral Research Associate, Department of Mechanical Engineering, RIT *Rochester, NY*

- Participated in curriculum development for the 2025 NSF-Engine Workforce Development Summer Camp for community college students.
- Designed and delivered a lecture on galvanic cells and their components, adapting content to accessibility to students without prior technical background.
- Developed instructional materials explaining battery classification (based on life cycle, format, electrode architecture, etc.) and operational principles.
- Collaborated with graduate students to integrate lecture content with hands-on experiments, ensuring theoretical concepts were reinforced through experiential learning.

Multiphysics Modeling for solid state battery **Sep 2024 – May 2025**
Postdoctoral Research Associate, Department of Mechanical Engineering, RIT *Rochester, NY*

- Developed and implemented a multiscale and multiphysics electro-chemo-mechanical framework (MOOSE + DEM) to quantify mechanically induced anisotropy in uniaxially pressed solid electrolytes, revealing direction-dependent Li transport and dendrite propagation mechanisms.

- Designed and experimentally validated an asymmetric ring-electrode architecture to redirect Li flux toward mechanically robust orientations (45°), increasing critical current density from 0.9 to 5.47 mA/cm^2 (5× enhancement).
- Established a quantitative relationship between catholyte effective mechanical properties (elastic modulus, compressive strength, toughness) and interfacial contact stability in composite cathodes using coupled continuum–granular simulations.
- Led microstructural engineering of sulfide catholytes via particle-size refinement (ball milling), reducing effective stiffness while preserving strength and improving electro-chemo-mechanical stability and full-cell capacity retention in ASSBs.

Quantum Field Expert and Outreach Specialist **Sep 2024 – Nov 2024**
Postdoctoral Research Associate, Department of Mechanical Engineering, RIT *Rochester, NY*

- Contributed as a quantum field expert at the event titled "Lecture: What Are Quantum Technologies?", organized by Quantum CT, a collaborative initiative between the University of Connecticut and Yale University.
- Played a key role in promoting quantum technologies and their impact on economic growth and societal advancement.
- Prepared and tailored content to educate diverse audiences, including educators, businesses, policymakers, and engineers, on the importance of quantum technologies and their future impact.
- For more details, visit: https://quantumct.org/events/quantum-up-challenge/panel-sessions/

Mesoscopic and Continuum Scale Modeling **Aug 2020 – Aug 2024**
Research Assistant–Funded by Air Force Research Lab (AFRL) and Industry *Storrs,CT*

- Explored diverse ferroelectric systems using Time Dependent-Landau Ginzburg Devonshire (TD- LGD) approach and phase-field modeling.
- Developed and applied physics based computational tool (in house built finite element software (based on C/C++) to simulate structure-property relationships in ferroelectrics materials.
- Modeled dielectric dispersion response in different ferroelectric systems utilizing an original approach developed at UConn and collaborating institutions.
- Installed and maintained an open-source finite element Multiphysics Object-Oriented Simulation Environment (MOOSE) and in-house built software–Ferret; performed calculations using an HPC environment.
- Extensively leveraged Python packages for both pre- and post-processing stages, developed and optimized high performance data workflow algorithm to perform efficient extraction and subsequent analysis of simulated data.
- Performed thorough statistical sampling and analysis across various post-processing stages, handling extensive datasets.
- Explored magnetic systems at the mesoscale using the Landau-Lifshitz-Gilbert (LLG) model.
- Developed, tested, validated, and applied the micromagnetic module developed in Ferret to study the properties of magnetic system.
- Participated in the development of a continuum-level solidification model. Analyzed and predicted thermal evolution of a system during solid-liquid phase transformation.
- Developed and explored functional properties, including electro-optical, magneto-optical, and elasto-optical responses in optical materials, utilizing Multiphysics coupling approach.

- Have experience in effectively working in a team and collaborating on a project spanning diverse research backgrounds, including atomistic modeling, metamaterial design, device fabrication, material growth, and characterization.
- Possess strong communication skills with extensive experience in creating presentations, delivering them at conferences, and authoring peer-reviewed journal papers.

Atomisitc modeling–(Density Functional Theory) **Aug 2018 – Aug 2020**
Research Assistant, Physics Department, UConn *Storrs, CT*

- Implemented a data-driven approach to derive the proxy formula for the electrostrictive tensor using thermodynamic potential.
- High throughput screening of materials from the Material Project database and employed Density Functional Theory (DFT) for robust calculations to identify materials with high electrostriction coefficients.

Atomistic Modeling–(Classical Molecular Dynamics) **Aug 2016 – July 2018**
Research Assistant, Physics Department, WSU *Dayton, OH*

- Employed classical molecular dynamics (MD) simulations to investigate the impact of hydrogen energy and surface temperature on hydrogen retention in a tungsten matrix within a nuclear fusion reactor.
- Focused on predicting the safety implications of hydrogen isotope accumulation, particularly tritium, for the overall reactor performance and operation.
- Investigated helium bubble formation and its influence on hydrogen trapping in a tungsten matrix at various surface orientations, aiming to predict and understand the durability and thermoelectric conversion efficiency of fusion reactors.

Presentations (Oral) and Conferences

- *"Mesoscale Finite Element Modeling of Diffusion Bonding Between Dissimilar Materials"*, Accelerated Discovery, Design, and Development of Materials for Extreme Environments, Washington State University, April 2026, Pullman, USA
- *"Finite Element Implementation of a Mesoscale Phase Field Formulation for High-Temperature Sintering of Polycrystalline Solids"*, Accelerated Discovery, Design, and Development of Materials for Extreme Environments, Washington State University, April 2026, Pullman, USA
- *"D-Wave Quantum: Understanding Quantum Technologies Webinar"*, QuantumUp! Challenge 2026, QuantumCT Webinar, March 2026, Virtual, USA
- *"Modeling the dielectric dispersion of* $BaTiO_3$ *with a local phase field method"*, Materials Science and Technology Conference, October 2023, Columbus OH, USA
- *"Modeling the relaxor dielectric dispersion of* $Ba_{1-x}Sr_xTiO_3$ *with a local phase field method"*, Materials Science and Technology Conference, October 2022, Pittsburg PA, USA
- *"Modeling the relaxor dielectric dispersion of with a local phase field method"*, Electronic Materials and Applications Conference, January 2022, Virtual participation, USA
- *"Predicting dielectric response in 5G materials with FEM simulations"*, Industrial affiliates program (IAP) annual meeting, July 2019, UConn, Storrs CT, USA.

Professional Memberships and Services

- Peer reviewer: Physical Review B, Physical Review Applied, ACS Applied Materials & Interfaces, Discover Materials
- American Physical Society
- American Ceramic Society (ACerS)
- Material Advantage
- Association for Iron and Steel Technology (AIST)

Workshops

- *"Accelerated Discovery, Design, and Development of Materials for Extreme Environments"*, April 2026, Pullman, USA.
- *"Moving ions with VASP"* Organized by VASP, Virtual, Nov 2023.
- *"MOOSE Workshop"* Organized by Idaho National Lab, Virtual, June 2023.
- *"Quantum Matter: Dynamics and Sensor"* jointly organized by the University of Connecticut, University of North Florida, and NORDITA, December 2022.
- *"CHiMaD Phase Field Workshop XIV"* organized by Center for Hierarchical Material Design, Evanston, IL, Hybrid, October 2022.
- *"CHiMaD Phase Field Workshop XII"* organized by Center for Hierarchical Material Design, Virtual, November 2021.
- *"IMS Quantum Matter Group 2019"* held at Los Alamos National Lab (LANL) as a visiting graduate student for the summer program 2019.

Honors Scholarships and Fellowships

- *"Modeling structure–properties relations in compositionally disordered relaxor dielectrics at the nanoscale"* paper **featured on the cover of Journal of Applied Physics** (2023).
- Conference Participation Award, UConn (2023).
- Doctoral Dissertation Fellowship, UConn (2023).
- Pre-Doctoral Dissertation Fellowship, UConn (2019).
- Guest Travel Fund, Los Alamos National Lab (LANL) (2019).

Leadership

- Senate, Graduate Student Senate, (2021-2023)
- Secretary, Nepalese Graduate Student Association, (2021-2022)
- Member, Student Welfare Committee, (2022-2023)
- Member, Graduate Employee and Postdoc Union (2018-2023)

References

- Dr. Scott Beckman, scott.beckman@wsu.edu, Professor, Phone: 509-335-8571, Acting Director, School of Mechanical and Materials Engineering, Washington State University, Pullman, WA, USA
- Dr. Sinisa Mesarovic, smesarovic@wsu.edu, Professor, School of Mechanical and Materials Engineering, Washington State University, Pullman, WA, USA,
- Dr. Serge Nakhmanson, serge.nakhmanson@uconn.edu, Associate Professor, Phone: 860-486-5252, Department of Materials Science and Engineering, University of Connecticut, Storrs, CT, USA